Duncan Rabenstein

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Education

University of British Columbia Kelowna, BC

Bachelor of Applied Science in Electrical Engineering - Mechatronics *September 2020 – May 2026*

- **Selected Coursework:** Signals and Communication Systems, Microcomputer Engineering, Digital Logic Design, Instrumentation & Data Analysis, Data Structures, & Technical Communication

Experience

Distribution Co-op Student May 2023 – December 2023

Primary Engineering and Construction *Kelowna, BC*

- Delivered technical customer support and project coordination for over 50 electrical and gas service installations, working directly with homeowners, electricians, and utility contractors
- Drafted detailed technical documentation, system layouts, and electrical schematics in AutoCAD according to municipal and utility engineering standards
- Accelerated gas service billing workflows by programming custom VBA automation scripts, eliminating manual data entry errors and cutting administrative processing time by 30%

Software Engineering Co-op Student January 2024 – April 2024

EarthDaily Analytics *Vancouver, BC*

- Collaborated within an Agile team to engineer Python-based software to identify homogeneous regions in Synthetic Aperture Radar (SAR) satellite imagery to enable automated processing of large datasets
- Automated SAR image processing workflows using the Science Toolbox Exploitation Platform and Linux scripts, reducing manual intervention and improving reproducibility of analysis pipelines
- Conducted research on methane emissions from thawing permafrost to support development of a software proposal

Projects

Embedded IoT Audio Controller | *ESP32, Embedded C++, UART, Python* June 2026 – July 2026

- Engineered a real-time signal processing engine in Python using Fast Fourier Transform (FFT) spectral analysis across 16 frequency bands to convert continuous audio data into low-latency visual commands
- Designed a custom 20-byte binary serial communication protocol with double-byte frame synchronization (`0xFF 0xFE`) to ensure noise resilience and eliminate packet corruption over serial links
- Programmed ESP32 microcontrollers in Embedded C++ to parse continuous UART serial telemetry streams at 50 FPS, executing deterministic real-time control loops to drive LED strips

Microwave Ice Detection Capstone Project | *C++, ESP32, RF Sensors* September 2025 – April 2026

- Architected a scalar network analyzer to evaluate and log the amplitude-frequency response of a resonant microwave sensor
- Utilized digital multimeters and oscilloscopes to perform signal calibration, hardware validation, and baseline sensor debugging
- Authored comprehensive technical documentation detailing the full design lifecycle, including system architecture schematics, component selection rationale, and final validation reports

Software Defined Radio M-PSK Testbed | *Python, GNU Radio, RF Modulation* January 2026 – April 2026

- Evaluated digital modulation schemes (4-PSK, 8-PSK, 16-PSK) by analyzing signal quality, frame synchronization, and carrier phase performance across simulated wireless channels

Technical Skills

Protocols & Networks: Serial (UART, RS-485 concepts), RF/Zigbee Concepts, TCP/IP Basics, & Wireless Synchronization
Troubleshooting & Tools: Digital Oscilloscope, Digital Multimeter (DMM), Function Generator, & Logic Analyzer
Software & Environments: Linux, Windows, Python, Embedded C/C++, AutoCAD, VBA, Git/GitHub, & VS Code
Soft Skills: Technical Support (Email/Phone), Technical Writing, Customer Coordination, & Field/Lab Testing

Interests

Ski Coaching, Rock Climbing, Mountain Biking, Soccer, Ultimate Frisbee, & Home Automation Hardware